Multi-arm Bandits

Ref. Sutton & Barto
Chapter 2.

* In RL, purely evaluative feedback indicates how good the action taken is, but not whether it is the best or the worst action possible.

* In this chapter, we study the evaluative aspect of Re-inforcement learning in a simplified setting: one that does not involve learning to act in more than one situation.

* n- Armed bandit problem:

You are faced repeatedly with a choice among n different options/actions. After each choice, you receive a numerical reward chosen from a stationary probability distribution that depends on the action you selected. Your objective is to maximize the expected total reward over some time period, for example, over 1000 action selections, or time steps.

* You do not know the action values with certainty, rather you know the estimates.

Treatment options

patients

Reward is the well-being

* Greedy action:

The action whose estimated value is the greatest.

* Exploitation: → Selecting the greedy action.

Exploration → otherwise.

* Exploitation is the right thing to do to maximize the expected reward in one step, but exploration may produce the greater total reward in the long term.

Action-value method:

True value of action $a \rightarrow q(a)$

Estimated value at the t^{th} time step $\rightarrow$ $Q_t(a)$.

If at the t^{th} time step action a has been chosen $N_t(a)$ times prior to t, yeilding rewards $R_1, R_2, \ldots R_{N_t(a)}$, then its value is estimated to be:

$$Q_t(a) = \frac{R_1 + R_2 + \ldots + R_{N_t(a)}}{N_t(a)}$$

When, $N_t(a) = 0$, then $Q_t(a) =$ some default value

As $N_t(a) \rightarrow \infty$, $Q_t(a) \rightarrow q(a)$ as per Law of large number.

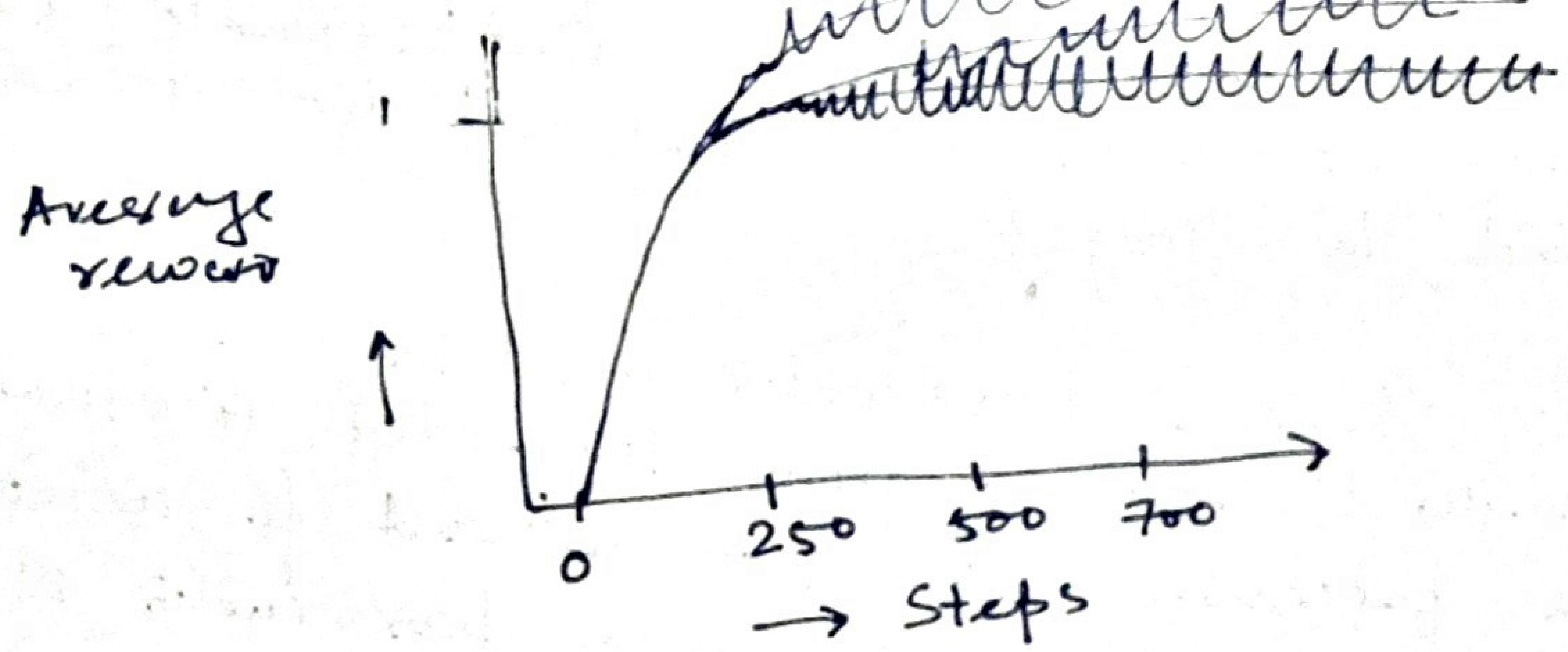
Greedy action selection:

$$A_t = \underset{a}{\arg\max}\ Q_t(a).$$

Where $\arg\max_a$ denotes the value of a at which the expansion that follows is maximized.

Ex: 10-Armed test bed:

Consider a set of 2000 randomly generati n-armed bandit tasks with $n = 10$. For each bandit, the action values $q(1), q(2), \ldots, q(10)$ were selected according to a Normal (Gaussian) distribution with mean 0 and variable 1. On the t^{th} time step with a given bandit, the actual reward R_t was the $q(A_t)$ for the bandit (where A_t was the action selected) plus a normally distributed noise term (mean $= 0$, variance $= 1$). Averaging over bandits, we can plot the performan & behaviour of various methods as they improve with experience over 1000 steps



Explore ε · greedy.

$$\epsilon = 0.1,\ 0.01$$

⊛ Incremental implementation

$$Q_t(a) = \frac{R_1 + R_2 + \dots + R_{N_t(a)}}{N_t(a)}$$

Here, R_1 to $R_{N_t(a)}$ need to be stored to compute $Q_t(a)$. The other method to compute is:-

$$Q_{K+1} = \frac{R_1 + R_2 + \dots + R_K}{K} = \frac{1}{K}\sum_{i=1}^{K} R_i$$

$$= \frac{1}{K}\left(R_K + \sum_{i=1}^{K-1} R_i\right)$$

$$= \frac{1}{K}\left(R_K + (K-1)Q_K\right)$$

$$\cancel{= \frac{1}{K}\left(R_K + (K-1)Q_K + Q_K - Q_K\right)}$$

$$= \frac{1}{K}\left(R_K + KQ_K - Q_K\right)$$

$$= Q_K + \frac{1}{K}\left(R_K - Q_K\right)$$

which holds even for $K=1$, i.e., $Q_2 = Q_1 + R_1 - Q_1$
$= R_1$

This update rule can be written in the form of:

$$\boxed{\text{NewEstimate} \leftarrow \text{OldEstimate} + \text{StepSize}\left[\text{Target} - \text{OldEstimate}\right]}$$

The expression [Target − Old Estimate] is an error in the estimate. It is reduced by taking a step toward the target.

- The stepsize parameter used in the incremental method changes from time-step to time-step. In processing the K^{th} reward for action a, that method uses a step size parameter of $\frac{1}{K}$

Nonstationary problem:

- A "non-stationary problem" in RL refers to an environment where the underlying dynamics like state transition probabilities or reward functions change over time, meaning the optimal action for a given state may not be the same throughout the learning process, posing a significant challenge for traditional RL algorithms that assume a stationary environment.

- In such cases, it makes sense to weight recent rewards more havily than long-past ones. One of the most popular ways of doing this is to use a constant step-size parameter.

$$Q_{K+1} = Q_K + \alpha [R_K - Q_K].$$

Where α is constant.

Now,

$$\begin{aligned}
Q_{K+1} &= Q_K + \alpha [R_K - Q_K] \\
&= \alpha R_K + (1-\alpha) Q_K \\
&= \alpha R_K + (1-\alpha)[\alpha R_{K-1} + (1-\alpha) Q_{K-1}] \\
&= \alpha R_K + (1-\alpha)\alpha R_{K-1} + (1-\alpha)^2 Q_{K-1} \\
&= \alpha R_K + \alpha(1-\alpha) R_{K-1} + (1-\alpha)^2\alpha . R_{K-2} \\
&= \alpha R_K + \alpha(1-\alpha) R_{K-1} + (1-\alpha)^2\alpha R_{K-2} + (1-\alpha)^3 \alpha R_{K-3} \\
&\quad \dots \alpha(1-\alpha)^{K-1} R_1 + \alpha(1-\alpha)^K Q_1
\end{aligned}$$

$$\Rightarrow Q_{K+1} = (1-\alpha)^K Q_1 + \sum_{i=1}^{K} \alpha(1-\alpha)^{K-i} R_i$$

How many rewards ago, i.e., $K-i$, the reward R_i was observed.

- Optimistic initial values;

Low initial value → the agent may be stuck to a particular action.

High initial value → More exploration initially.

——— x ———